SOAP NOTE GENERATOR

Presented By

Mukta Naik	612203122
Sujal Nandapurkar	612203123
Revati Patare	612203131
Tanuja Patil	612203138

PROBLEM STATEMENT

Generating structured SOAP notes from medical conversations is a challenging NLP task due to unstructured dialogue, multiple speakers and contextual medical information. Most existing clinical NLP systems focus on English datasets, with limited support for Indic languages such as Marathi and Hindi. This project develops an automated pipeline that performs speech-to-text conversion, extracts clinical entities from conversations, and generates SOAP notes in Indic languages.

The system aims to:

- Process medical dialogue transcripts in indic languages
- Perform clinical entity extraction and contextual understanding
- Classify relevant information into Subjective, Objective, Assessment and Plan sections
- Generate structured SOAP notes that can assist in automated clinical documentation within healthcare systems

OBJECTIVES

1. **Dataset Expansion and Diversity**
   To improve dataset diversity using translation pipelines and LLaMA-generated synthetic conversations.

2. **Automated SOAP Note Generation**
   To architect an NLP engine that synthesizes the transcribed data into SOAP note in the supported language.

3. **Handling 'Code-Switching'**
   To handle code-mixed medical conversations where speakers switch between languages.

4. **Cross-Lingual Translation & Mapping**
   To develop a pipeline that converts Indic language complaints into standardized medical form.

5. **Clinical Entity Recognition**
   To extract key medical entities such as symptoms, emotions, duration, and severity from conversational data.

6. **Multilingual Speech Recognition (ASR)**
   To develop a multilingual ASR system capable of transcribing English and Indic languages with high accuracy.

7. **Multi-Speaker Diarization**
   To use speaker diarization to differentiate between doctor and patient dialogue for accurate SOAP section classification.

8. **Validation Framework**
   To evaluate generated SOAP notes for clinical relevance, accuracy and structural completeness.

DATASET DESCRIPTION

OVERVIEW


```
{
  "session_id": 328,
  "phq8_score": 4,
  "severity": "minimal",
  "gender": "male",
  "split": "train",
  "total_turns": 238,
  "dialects": {
    "standard_pune": [
      {
        "role": "Doctor",
        "text_en": "hi i'm ellie thanks for coming in today",
        "text": "हाय मी एली आहे आज आल्याबद्दल धन्यवाद"
      },
      {
        "role": "Doctor",
        "text_en": "i was created to talk to people in a safe and secure environment",
        "text": "मी सुरक्षित आणि सुरक्षित वातावरणात लोकांशी बोलण्यासाठी तयार केले आहे"
      },
      {
        "role": "Doctor",
        "text_en": "i'm not a therapist but i'm here to learn about people and would love to learn about you i'll ask a few questions to get us started",
        "text": "मी एक थेरपिस्ट नाही पण मी इथे लोकांबद्दल जाणून घेण्यासाठी आलो आहे आणि तुमच्याबद्दल जाणून घ्यायला मला आवडेल मला सुरुवात करण्यासाठी काही प्रश्न विचारतील"
      },
      {
        "role": "Doctor",
        "text_en": "and please feel free to tell me anything your answers are totally confidential",
        "text": "आणि कृपया मला काहीही सांगा तुमची उत्तरे पूर्णपणे गोपनीय आहेत"
      },
      {
        "role": "Doctor",
        "text_en": "are you okay with this",
        "text": "तुला हे ठीक आहे का?"
      },
      {
        "role": "Patient",
        "text_en": "yes",
        "text": "होय"
      },
      {
        "role": "Doctor",
        "text_en": "okay",
        "text": "ठीक आहे"
      },
```


DATASET SOURCE

The dataset is derived from the DAIC-WOZ mental health interview corpus, which contains therapist–patient conversations.

DATA PROCESSING

Parsed DAIC-WOZ transcripts to obtain conversation turns, speaker roles, and session metadata such as PHQ-8 score, severity, and gender and adapted the conversations into multiple Marathi dialect styles.

TRANSLATION & EXPANSION

English conversations were translated into Marathi using a custom translation pipeline. The dataset was further expanded using the LLaMA model to generate additional synthetic conversations while preserving clinical context and PHQ-8 labels.

CHALLENGES

Dataset Language Limitation

The DAIC-WOZ dataset is originally in English, while our system focuses on Marathi conversational data, requiring additional translation and preprocessing.

Maintaining Clinical Context During Translation

Direct translation sometimes produced incorrect or unnatural medical terminology, making it difficult to preserve the clinical meaning of conversations.

Dialect Variations in Marathi

Marathi has several dialects such as Pune, Konkan, Vidarbha, and Marathwada, which introduce variation in vocabulary and sentence structure.

Entity Extraction in Marathi

Most existing NER models are designed for English, so identifying medical entities in Marathi conversations required additional rule-based patterns.

Dataset Expansion

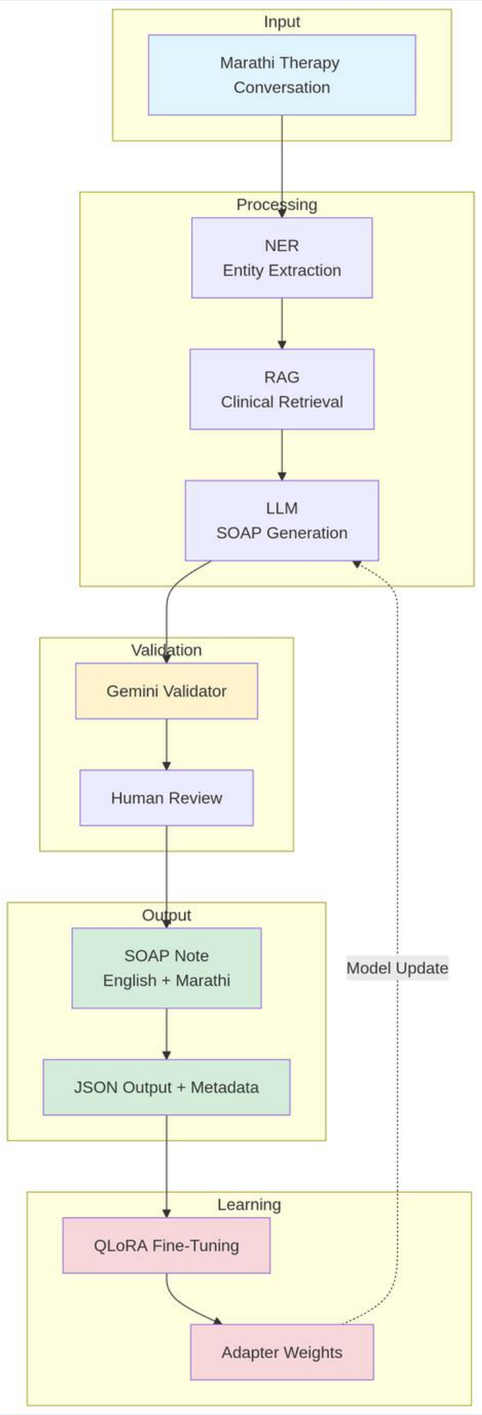
Synthetic conversations generated using LLaMA sometimes produced hallucinated or clinically inconsistent information, requiring additional validation to maintain dataset quality.

Generating Clinically Accurate SOAP Notes

Ensuring that generated SOAP notes were structured, concise, and medically relevant required careful prompt design and validation.

SYSTEM ARCHITECTURE

The proposed framework converts unstructured Marathi medical conversations into structured SOAP notes using a modular NLP pipeline. It combines entity extraction, contextual retrieval, language modeling, validation, and model learning to generate clinically meaningful documentation.



NAMED ENTITY RECOGNITION

The NER module extracts clinically relevant information from therapy conversations. A hybrid approach combining pretrained biomedical NER models, IndicNER model and rule-based keyword matching is used to identify mental health related entities from Marathi and English conversational text.

NER aims to :

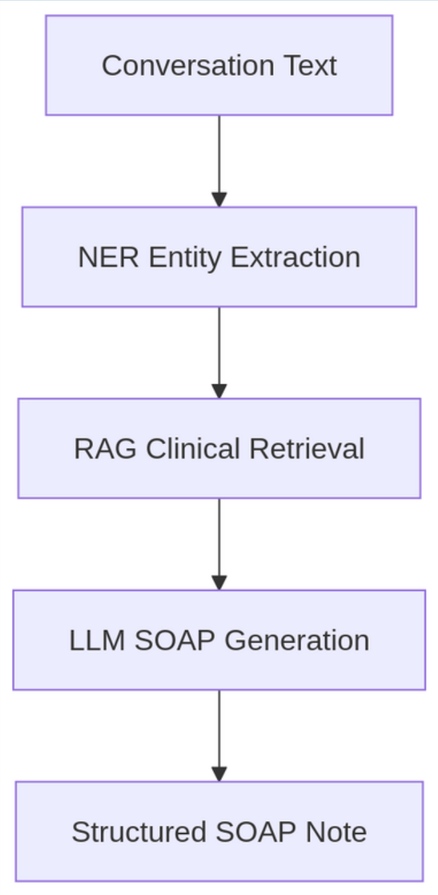
- Extract medical entities from conversational therapy transcripts
- Use **BioBERT / IndicNER** models for biomedical entity recognition
- Apply **rule-based keyword matching** for mental health terminology
- Handle code-mixed Marathi–English conversations
- Provide structured entities used by the RAG and LLM modules

SOAP NOTE GENERATION

The system uses a Retrieval-Augmented Generation (RAG) framework combined with a Large Language Model to generate clinically accurate SOAP notes. Clinical knowledge retrieved from a vector database is provided as contextual information to the LLM, improving the accuracy and reliability of generated medical documentation.

Key Functions :

- RAG retrieves relevant clinical terminology from a vector database
- Uses ChromaDB vector similarity search to find related medical concepts
- Retrieved clinical context is passed to the LLM
- The LLM generates structured SOAP notes based on conversation data and retrieved knowledge
- Improves medical accuracy and contextual understanding



QLORA FINE TUNING

LoRA Adapter Layers

Instead of modifying all model parameters, low-rank adapter matrices are inserted into attention layers (q_proj, k_proj, v_proj, o_proj) to learn task-specific behavior.

Prompt-Response Training

Training data is formatted as prompt + response pairs, where therapy conversation inputs are mapped to structured SOAP note outputs.

Parameter Efficient Learning

Only the LoRA adapter weights are updated during training, while the base model weights remain frozen.

4-bit Quantization

The base LLM is loaded using 4-bit precision, significantly reducing memory usage and enabling efficient training on standard GPUs.

ARCHITECTURE AND TRAINING PIPELINE

39 M
Parameters (Whisper-tiny)

448
Training clips (aug. from 112)

20 %
English WER (medical test set)

V 4
Inference version (halluc. suppressed)

↓ Hallucination Filter + Text Cleaning → **Timestamped Speaker**-Labelled Transcript

Whisper-tiny — Model Specs	
Parameters	39M (smallest Whisper variant)
Encoder	4× Transformer layers, d_model=384
Decoder	4× Transformer layers, vocab=51,865
Input	80-band Mel spectrogram, ≤30s
Tokenizer	BPE — supports Devanagari script
Framework	HuggingFace Transformers + Seq2SeqTrainer

Training Pipeline — 5 Steps

1. **Generate synthetic data**
   Google TTS → 140 Marathi+EN medical sentences
2. **Prepare & validate**
   16kHz mono resample, silence trim, 80/10/10 split
3. **Data augmentation ×4**
   Speed, pitch, noise, volume — 112 → 448 clips
4. **Fine-tune Whisper**
   Seq2SeqTrainer, cross-entropy loss, early stopping
5. **Evaluate + quantize**
   WER/CER per language, INT8 → 144MB to 81MB

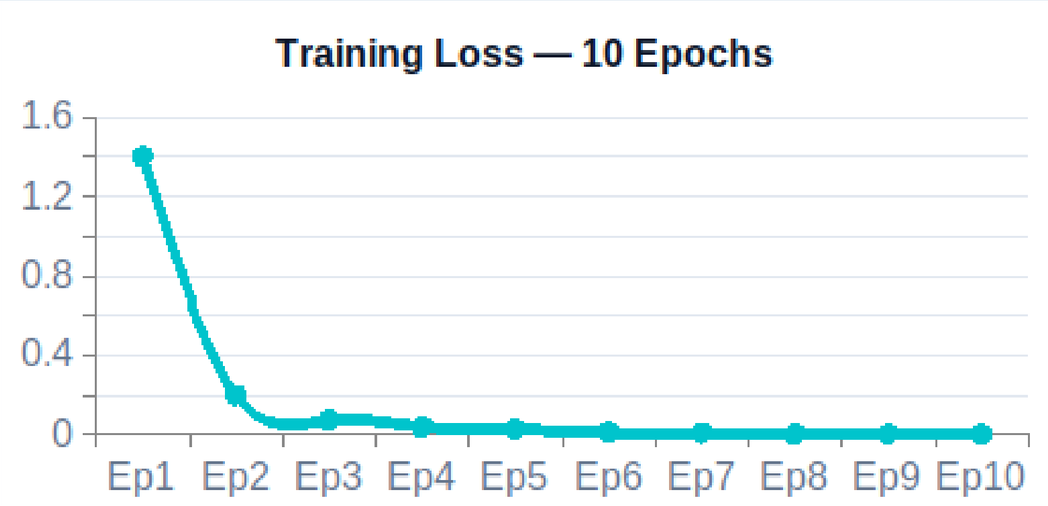
TRAINING RESULTS & HALLUCINATION CONTROL

⚠ What is Whisper Hallucination?

On silent/noisy chunks, Whisper generates plausible-sounding text instead of nothing — looping high-probability phrases indefinitely. A known failure mode of all autoregressive models.

Output — 305_AUDIO.wav (v1 inference):

```
[04:03]  "I feel like being outdoors. I feel like being outdoors..."  ×32
[17:33]  "having a feeling of having a feeling of having a feeling..."  ×50
```



3-Layer Fix (v4)

① VAD

Skip chunks where <10% of 20ms frames have speech energy — silent segments never reach Whisper

② Generation penalties

no_repeat_ngram_size = 3 blocks 3-word loops
repetition_penalty = 1.2 penalises repeated tokens

③ Post-filter

Unique word ratio <25% → discard · Known hallucination phrase blacklist · Sentence deduplication

Component	Status	Current Metric	Next Step
Data Pipeline + Augmentation	✅ **Complete**	448 clips (4× from 112)	Add real IndicVoices audio
Whisper Fine-tuning (EN+MR)	✅ **Complete**	Train loss 1.40 → 0.0003	Retrain with real MR speech
English Medical ASR	✅ **Working**	WER: 20%	Target <8% with GPU + data
Marathi ASR	🔄 **Improving**	WER: 38.7% — data mismatch	EkaCare + IndicVoices-R data
Hallucination Control	🔄 **Improving**	VAD + penalties + post-filter	Tune on real recordings

Root Cause — High Marathi WER

Training used gTTS synthetic voice — robotic, flat, no background noise. Real doctor-patient speech has natural intonation, fast speech and overlapping voices. The model learned the right words but the wrong acoustic patterns.

Fix: EkaCare clinical conversations + AI4Bharat IndicVoices-R (1,704 hrs, 10,496 real speakers).

Expected WER after retrain: <25% Marathi, <8% English.

Key Technical Contributions

Multilingual fine-tuning

First Whisper fine-tune for MR-EN code-switched medical speech

Data augmentation pipeline

8 transforms, 4× dataset multiplication, no extra recordings

Custom VAD

RMS + 20ms frame analysis, no external library, no ffmpeg

Hallucination suppression

3-layer: generation penalties + unique ratio + phrase blacklist

Production inference

28-min audio tested, speaker hints, .txt+.json output

Full evaluation framework

WER/CER per language, examples.csv for targeted improvement

REFERENCES

1.PHQ-8 DEPRESSION QUESTIONNAIRE

Patient Health Questionnaire-8 (PHQ-8) is a validated tool used to measure depressive symptoms and severity in clinical studies.
https://selfmanagementresource.com/wpcontent/uploads/English_-_PHQ-8-1.pdf

2.DAIC-WOZ DATASET

Distress Analysis Interview Corpus – Wizard of Oz dataset used for depression analysis from clinical interviews.
https://dcapswoz.ict.usc.edu/wwwdaicwoz/

3.AUTOMATIC SOAP NOTE GENERATION – COEP TECHNOLOGICAL UNIVERSITY

4.AI4BHARAT – INDIC NLP RESOURCES

AI4Bharat is an open research initiative from IIT Madras focused on developing open-source datasets, models and tools for Indian languages. It provides multilingual NLP resources such as IndicNER and translation models supporting multiple Indic languages.
https://ai4bharat.iitm.ac.in/

5. EMOINHINDI

Singh, G.V., Priya, P., Firdaus, M., Ekbal, A., & Bhattacharyya, P. (2022)
EmoInHindi: A Multi-label Emotion and Intensity Annotated Dataset in Hindi for Emotion Recognition in Dialogues.
https://github.com/TusharD-047/EmoInHindi

CONCLUSION

This project presents a **modular NLP pipeline for automated SOAP note generation from Marathi therapy conversations.** By combining speech processing, entity extraction, contextual retrieval, and large language models, the system transforms unstructured conversational data into structured clinical documentation.

The proposed architecture integrates **Named Entity Recognition (NER), Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs)** to accurately identify medical entities, retrieve relevant clinical terminology, and generate structured SOAP notes. The system further incorporates validation mechanisms and **QLoRA-based fine-tuning** to improve model reliability and adaptability.

Experimental results demonstrate that the pipeline can effectively capture **clinical information from conversational text and convert it into standardized medical documentation,** supporting automated healthcare record generation.

Overall, the proposed framework contributes toward **AI-assisted clinical documentation for Indic languages**, helping reduce manual workload for healthcare professionals while improving accessibility of medical NLP systems for regional languages.

THANK YOU VERY MUCH!